

L3 Aluminum and Composite Tensile Testing

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All information documented and discussed in the following report is under the sole authorship of Anuranan Bharadwaj, Kalkamanali Satvaldy, Ryan Orr, James Nguyen. No part of this report has been published or submitted for publication.

Signed:

KS, AB
Amr, JN

Abstract

Tensile Testing is an important process required to determine the mechanical behavior of a material. This helps engineers maintain a balance between cost, weight, and safety when using such materials. Two of the most common materials are lightweight aluminum alloys and glass/carbon fiber reinforced polymer composites (GFRP and CFRP), chosen for their cost, mechanical performance when compared to their weight, and a myriad of other factors. Tensile testing is the act of slowly applying tensile stress to a sample of the chosen material, often a rod made of the material, until critical failure. From this, stress and strain are noted and plotted, allowing the calculation of other mechanical properties such as Young's Modulus (E). In addition to this, metallography can be conducted, a method that allows one to examine microstructures in a metal using a microscope, for example, the microstructures caused when a material fractures. This is what our experiment consisted of: placing three different test samples, Aluminum-2024, CFRP, and GFRP, into a Tinius Olsen LVDT Extensometer and having them placed under load until fracturing. Plotting this gives us a Young's Modulus for the Aluminum, CFRP, and GFRP of 693 MPa, 1118.6 MPa, and 373.9 MPa, and an ultimate stress of 497 MPa, 1030 MPa, and 469 MPa, respectively. The Aluminum sample showed a yield stress of 361 MPa. Comparing these values to their actual values gives us a percent difference for the Aluminum, CFRP, and GFRP, of we see that most values are below 3% except for the ultimate stress on CFRP of around 8% and the Young's Modulus of around 50%. In an effort to validate our values, we estimated Young's Modulus of the CFRP sample using the rule of mixtures and observing the microstructures of the sample under a microscope, giving us an estimated Young's Modulus of 126984 MPa, which is 6% different when compared to the published value. We also went on to find the weight-specific strength finding that Aluminum-2024 had the lowest value, sitting at around 178777 MPa*mm³/g, followed closely by GFRP at around 250482 MPa*mm³/g, and both of these values being easily surpassed by CFRP at 717297 MPa*mm³/g.

Introduction

Tensile testing is one of the most basic techniques for describing the mechanical properties of materials in engineering. This test provides important parameters of the material, such as Young's modulus, yield strength (if yielding), ultimate strength, and ductility, from the resulting stress-strain curve. The test involves continually measuring stress and strain as a specimen is exposed to an increasing uniaxial load until it fractures.

Modern tensile testing evolved with the development of the universal testing machine, which allows precise control of load and strain. Tensile testing continues to be the industry standard for materials research and quality control in aerospace and other cutting-edge sectors.

For this lab, aluminum alloys and fiber-reinforced composites are used to examine the mechanical properties. Many aerospace applications use aluminum alloys like 6061-T651, 7075-T651, and 2024-T351, as well as fiber-reinforced composites such as CFRP (carbon-fiber reinforced polymer) and GFRP (glass-fiber reinforced polymer) because of their excellent weight-specific mechanical performance, good corrosion resistance, affordability, and fabrication know-how.

Composites are multiphase materials that combine different properties to produce a superior material with a better combination of properties. The strongest composites contain long continuous fibers made from very stiff and strong (but usually brittle) materials like glass, carbon, etc. They are surrounded by a more ductile matrix such as aluminum or epoxy. The overall mechanical response can be predicted using the rule of mixtures:

$$\sigma_c = \sigma_m V_m + \sigma_f V_f$$

$$E_c = E_m V_m + E_f V_f$$

where,

σ_c – matrix stress (MPa)

V_m – matrix volume fraction

σ_f – fiber stress (MPa)

V_f – fiber volume fraction

E_m – matrix Young's modulus (GPa)

E_f – fiber Young's modulus (GPa)

Axial stress in a conventional beam can be calculated using:

$$\sigma = \frac{My}{I}$$

where,

M – bending moment (N·mm)

y – distance from the neutral axis (mm)

I – second moment of area about the neutral axis (mm^4)

Strain is measured as the ratio of elongation to original length:

$$\varepsilon = \frac{\Delta l}{l_0}$$

where,

Δl – change in length (mm)

l_0 – original length (mm)

These two quantities are related through Hooke's Law in the elastic region:

$$E = \frac{\sigma}{\varepsilon}$$

An extensometer is used to measure linear strain precisely during the test, particularly during the first elastic period where deformation is minimal. In order to observe plastic deformation and fracture behavior, the test is conducted beyond the yield point. Aluminum alloys display yielding behavior before necking, while the composites tend to fracture in a brittle manner.

This experiment aims to compare the tensile behavior of ductile and brittle structural materials, including aluminum alloy, GFRP, and CFRP. Additionally, optical microscopy of polished samples will give information on the microstructural factors influencing their mechanical performance.

Equipment/Apparatus

- 2024-T351 aluminum tensile specimen
- GFRP tensile specimen (polyester matrix)
- CFRP tensile specimen and small CFRP polishing samples
- 6061-T6 aluminum gripping tabs/tubes
- Calipers
- Mass scale
- Permanent ink marker
- 240 grit SiC sandpaper
- Table protection (tarp, construction paper, etc.)
- Wipes
- Gloves
- Acetone
- Cyanoacrylate adhesive
- Tinius Olsen 150ST Electromechanical Universal Testing Machine with computer/software
- Tinius Olsen LVDT Extensometer
- Hot glue and gun
- Hand sander (with 240, 320, 400, and 600 grit SiC sandpaper)
- Grinding and polishing machine (with 320 grit SiC sandpaper and soft polishing cloth)
- 1 μ deagglomerated alpha alumina (aluminum oxide slurry)
- Polishing sample holder
- Water
- Methyl alcohol
- Wall-mounted hand dryer
- Motic metallurgical microscope with illuminator, digital camera, Whibal card, imaging computer/software, and abbreviated imaging instructions

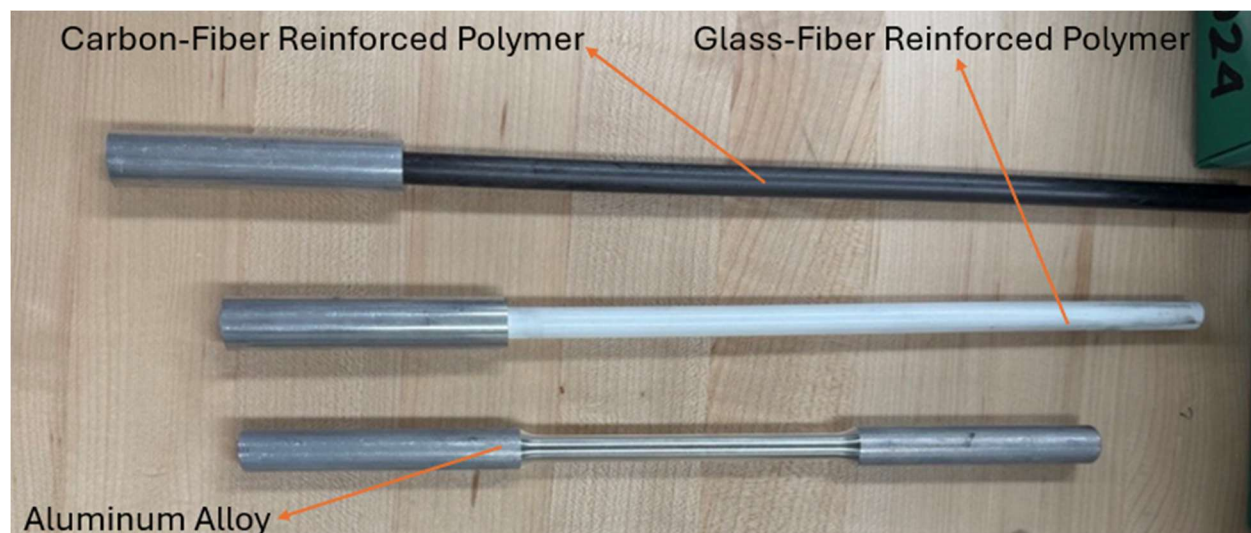


Figure 1: Specimens



Figure 2: Specimens after tensile testing



Figure 3: Tinius Olsen 150ST Electromechanical Universal Testing Machine

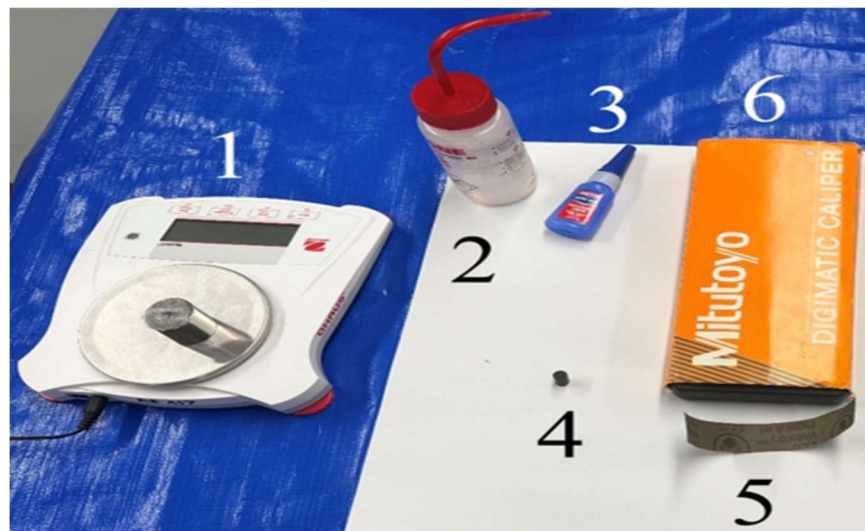


Figure 4: Experimental equipment used - (1) Mass Scale, (2) Acetone, (3) Cyanoacrylate Adhesive, (4) Small CFRP polishing sample, (5) 240 grit SiC sandpaper, (6) Calipers



Figure 5: Water Lubricated Sander



Figure 6: Grinding and Polishing Machine



Figure 7: Hot Glue Gun



Figure 8: Hand Dryer

Method

A. Tensile Sample Preparation and Gripping Tube Installation

1. Lightly sanded the gauge sections of the Aluminum Alloy, GFRP, and CFRP specimens using 240-grit sandpaper.
2. Measured the diameter of each specimen at five different positions near the center using digital calipers and calculated the mean value.
3. The overall length of GFRP and CFRP specimens was measured, and the mass was recorded using a digital balance.
4. Ensure that the GFRP and CFRP fit reasonably snugly into the Aluminum alloy gripping tubes. If the gripping tubes are snug or slightly loose, about 75 mm of gripping distance from each end was sanded using 240-grit SiC sandpaper for better adhesion.
5. If too tight, use a grinding and polishing machine and uniformly sand using water lubricated 320 grit SiC sandpaper.
6. Sharp inner edges on the gripping tubes were rounded over using 240 grit sandpaper to reduce stress concentrations.
7. The specimens and gripping tubes were cleaned, rinsed with acetone, and left to dry.
8. Few drops of cyanoacrylate adhesive were applied inside the gripping tubes. Gripping tubes were placed vertically on the table, lined up to the sample, and then quickly rotated and pressed the sample all the way through the hole.
9. Plug in the hot glue gun and let it warm up

B. Testing Machine Setup

1. Turn ON the circuit breaker
2. The rotary switch on the left bottom of the Tinius Olsen 150ST Testing Machine was rotated clockwise to the ON position until beep sound confirmations occurred.
3. The computer and monitor were turned ON, and the TOVMC software was started.
4. Speeds were checked so that they were:
 - Speed 1 (0.1 mm/min)
 - Speed 2 (10 mm/min)
 - Jog Up/Down (300 mm/min)
5. The Horizon software was launched.

C. Hydraulic Gripping Unit Setup

1. The MITS hydraulic grip supply was turned ON.
2. Grip faces were checked to see if they were seated correctly.
3. Gripping pressure of 4000 psi was used for all specimens

D. Tensile Test

1. The following order for the test was used: 1) Aluminum Alloy, 2) GFRP, and 3) CFRP

2. For the Aluminum Alloy, the “ERAU Metal Tensile” test was loaded and “ERAU Composite Tensile” for the GFRP and CFRP.
3. The load cell (Channel 1) was zeroed without applying any load by pushing the correct button.
4. The extensometer was checked to see if it was operating correctly.
5. The machine was positioned to insert the specimen into the lower grip using the job controls, and a clamping pressure of 4000 psi was applied.
6. The machine was further jogged down to grip the upper end of the sample and the 4000 psi grip was applied again.
7. The grip position was adjusted using speed 1 to bring the force shown in the extensometer close to zero.
8. To fix the proper gage length, the extensometer zeroing pin was set. It was spun so the rounded side touched the lever arm.
9. Using the centering groove and checking the vertical alignment, the extensometer was attached near the specimen center in a way that it wouldn't hit the grips.
10. A small amount of hot glue was applied to where the knife edges were in contact with the test specimen, and the strain and position (Channel 3 and Channel 2, respectively) were zeroed.
11. The calculated cross-sectional area was then entered into the software.
12. The safety shield was then attached, and a green arrow was clicked to automatically run the test (5 mm/min for metal and 1 mm/min for composite).

E. Saving the Data and Removing the Extensometer

1. Test data was highlighted using a mouse.
2. Data was exported to a USB drive and renamed to a clear a sensible name, e.g., Al 2024-T351 Tensile Test.xls.
3. The extensometer was removed, and the test specimen was un-gripped.
4. The equipment was then turned off by rotating the switch on the Tinius Olsen 150ST Testing Machine counterclockwise and turning the circuit breaker to the OFF position.

F. Metallography Specimen Preparation

1. A small CFRP polishing sample was polished using a hand sander, water lubricant, and finer water-lubricated sandpaper, starting with 240 grit, then progressing to 320, 400, and 600 grit SiC to achieve a flat surface on one side of the sample.
2. The sample's edges were then chamfered using a grinding machine with water-lubricated 320-grit SiC sandpaper.
3. The sanded surface of the sample was then polished with a polishing machine, soft cloth, and a polishing slurry with a 1:3 ratio of 1 μ deagglomerated alpha alumina and with water to achieve a mirror-like finish.
4. The sample was rinsed with tap water and with methyl alcohol while wearing gloves.

5. The sample was dried using the wall mounted hand dryer

G. Optical Microscopy Setup and Procedure

1. The metallurgical microscope and the microscope light were turned ON.
2. The microscope light was then adjusted to its level by turning the knob.
3. The trinocular level was located and pulled fully out to allow half of the light to enter the camera.
4. The monitor and the computer were turned ON, and students were logged in to start the Images Plus software.
5. Abbreviated imaging instructions were followed that were located near the microscope.
6. Multiple locations at different magnifications of the polished surface were investigated.
7. At least 5 images at different locations and magnifications were saved to a USB drive.
8. The equipment was turned OFF in the reverse order.
9. The work station was thoroughly cleaned up.

Results

One of the first steps of the experiment was to measure the diameter, length, and mass of the Aluminum Alloy 2024-T351, Carbon Fiber, and Glass Fiber samples, as shown in the table below:

Sample	Parameter	1	2	3	4	5	Mean
2024-T351	d (mm)	8.68	8.53	8.67	8.55	8.53	8.58
GFRP	d (mm)	9.53	9.53	9.53	9.53	9.53	9.53
	L (mm)	295					
	m (g)	39.4					
CFRP	d (mm)	9.37	9.40	9.39	9.36	9.40	9.38
	L (mm)	320					
	g (mm)	31.78					

Table 1: Geometric Parameters of specimens

Pictures taken from the microscope of the CFRP sample shown below in Figures 9-13:

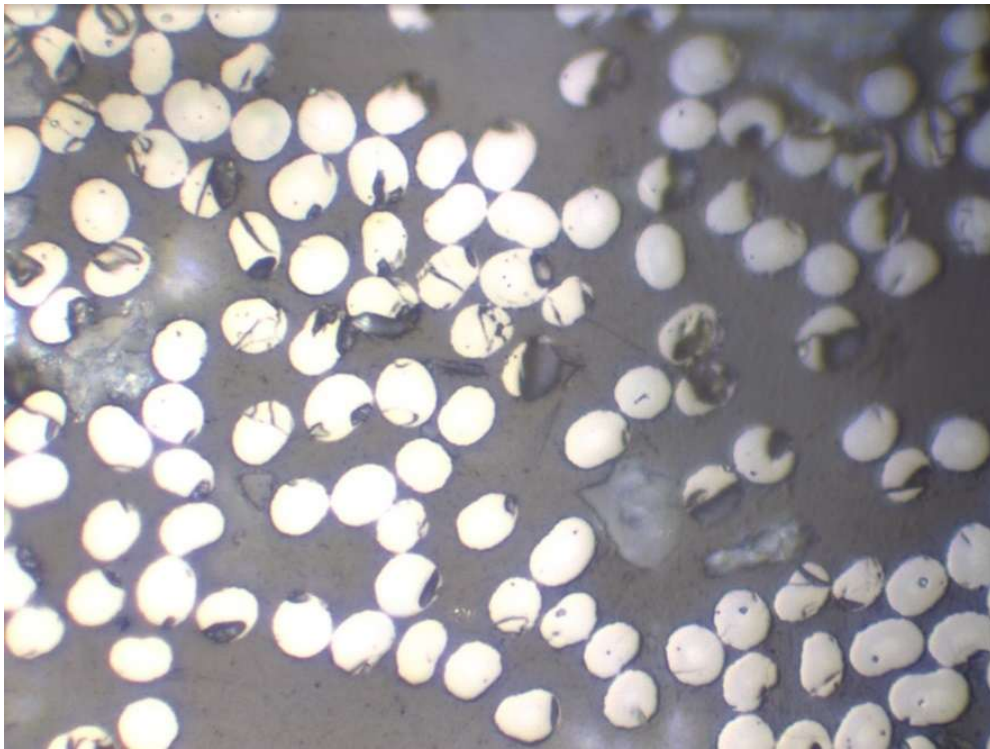


Figure 9: Zoomed picture number 1 of CFRP fibers

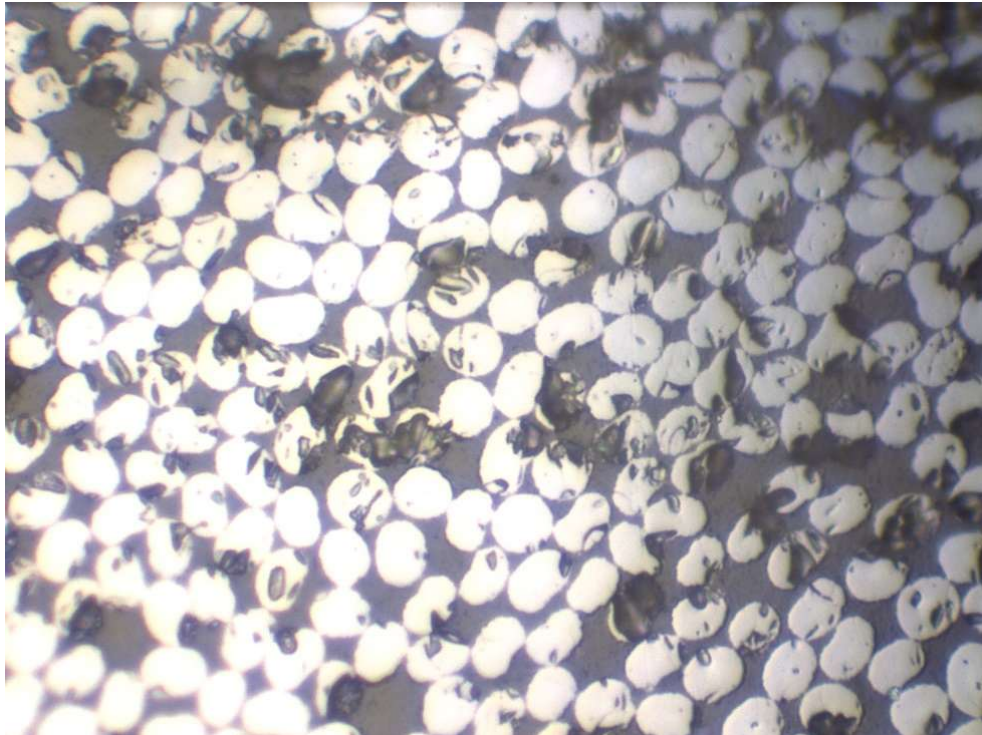


Figure 10: Zoomed picture number 2 of CFRP fibers

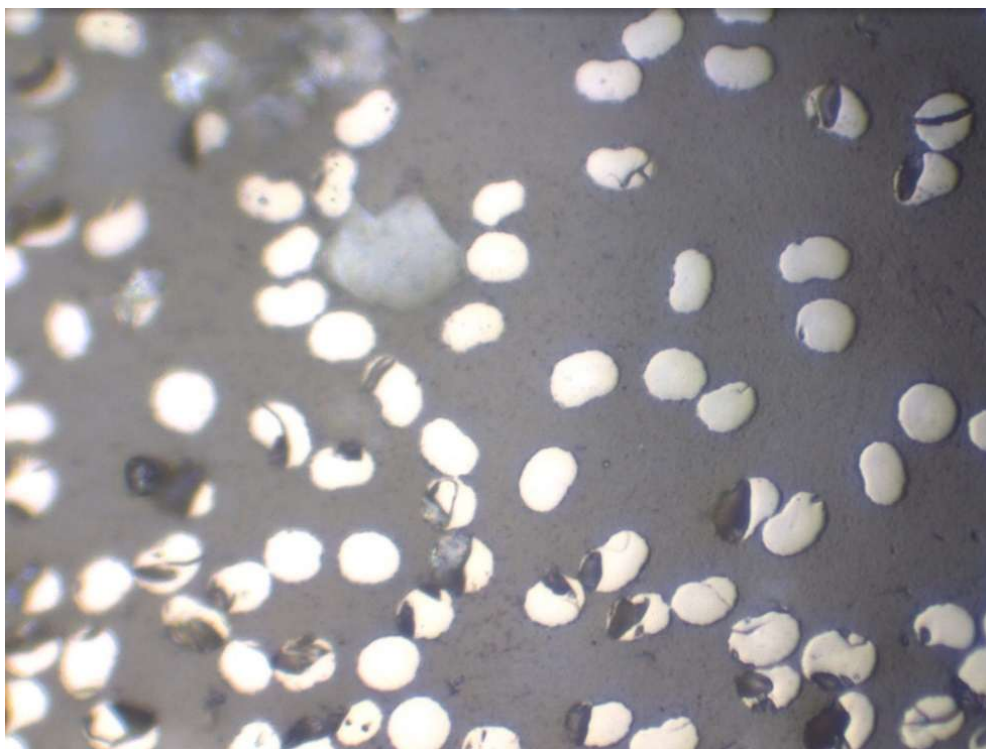


Figure 11: Zoomed picture number 3 of CFRP fibers

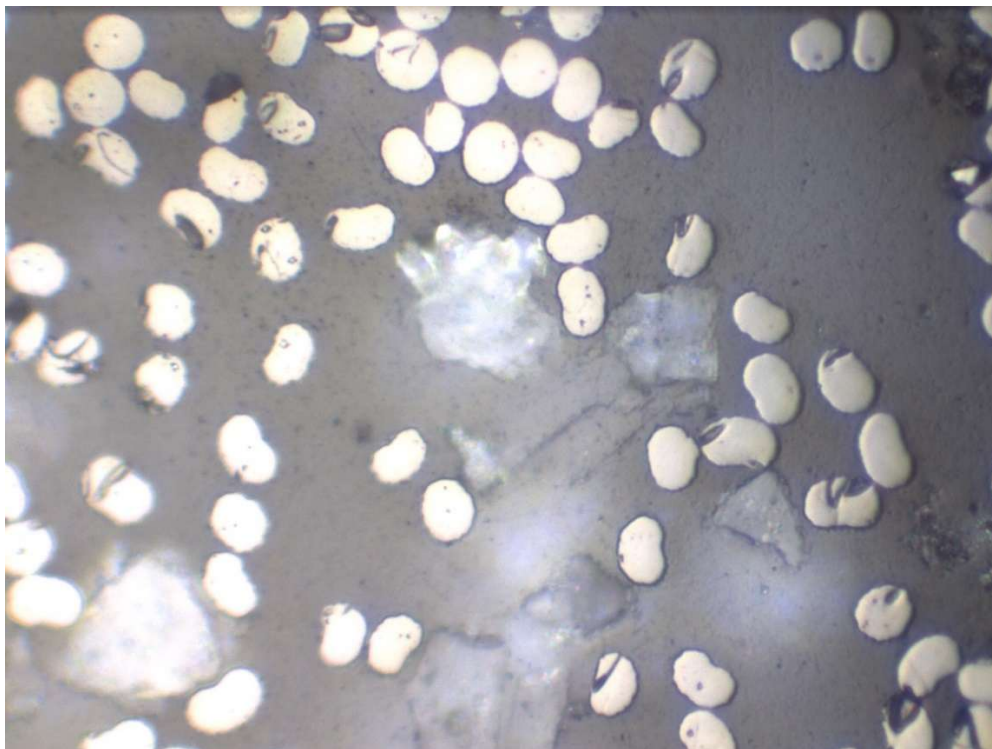


Figure 12: Zoomed picture number 4 of CFRP fibers

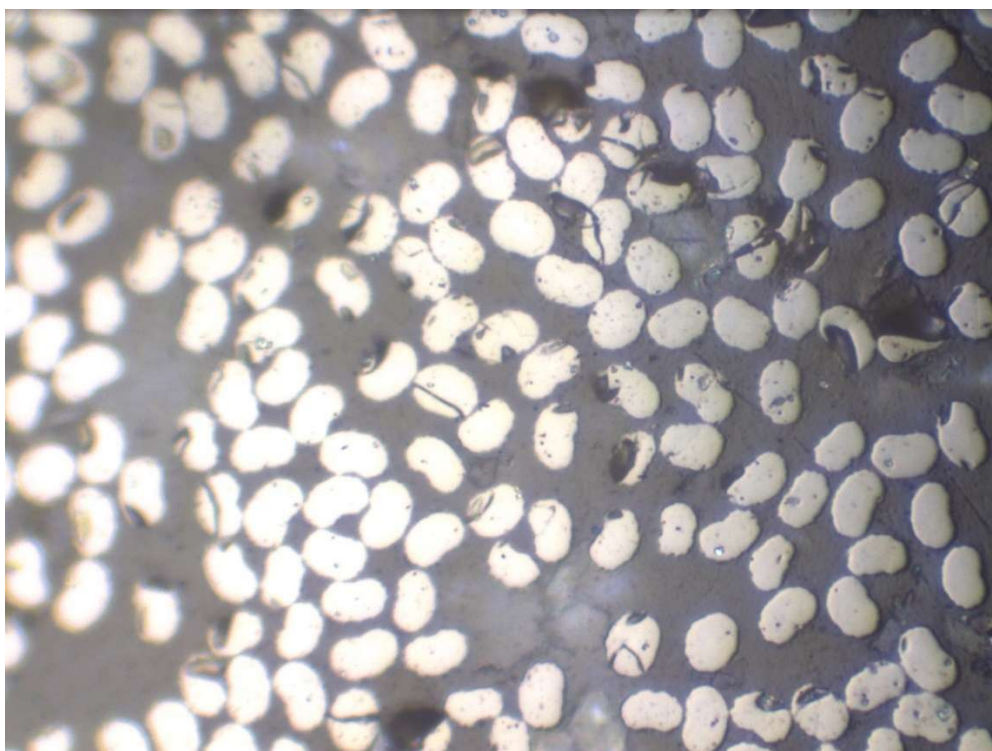


Figure 13: Zoomed picture number 5 of CFRP fibers

Analysis

Before the analysis of each test, the data were recorded on a stress(σ) vs strain(ϵ) curve. In each curve, there is a point where the curve stopped being predictable and started to regress, this is the point where it either started to neck or broke its fibers. Through this curve and associated data, the Young's modulus (E), ultimate stress (σ_u), and yield stress (σ_y) (only for the aluminum beam) could be found.

When calculating E , 5 different locations on the elastic curve were calculated using equation 1, then averaged to find a mean value. This produced an average of 693 MPa for Al 2024, 373.9 MPa for GRFP, and 1118.6 MPa for CRFP.

$$E = \sigma/\epsilon (1)$$

σ_u was found by finding the maximum value of each data point. This resulted in 497 MPa for Al 2024, 469 MPa for GFRP, and 1030 MPa for CFRP. As Al 2024 is a ductile metal, it also has σ_y . This was found by finding the 0.2 offset for the strain values, giving a σ_y of 361 MPa.

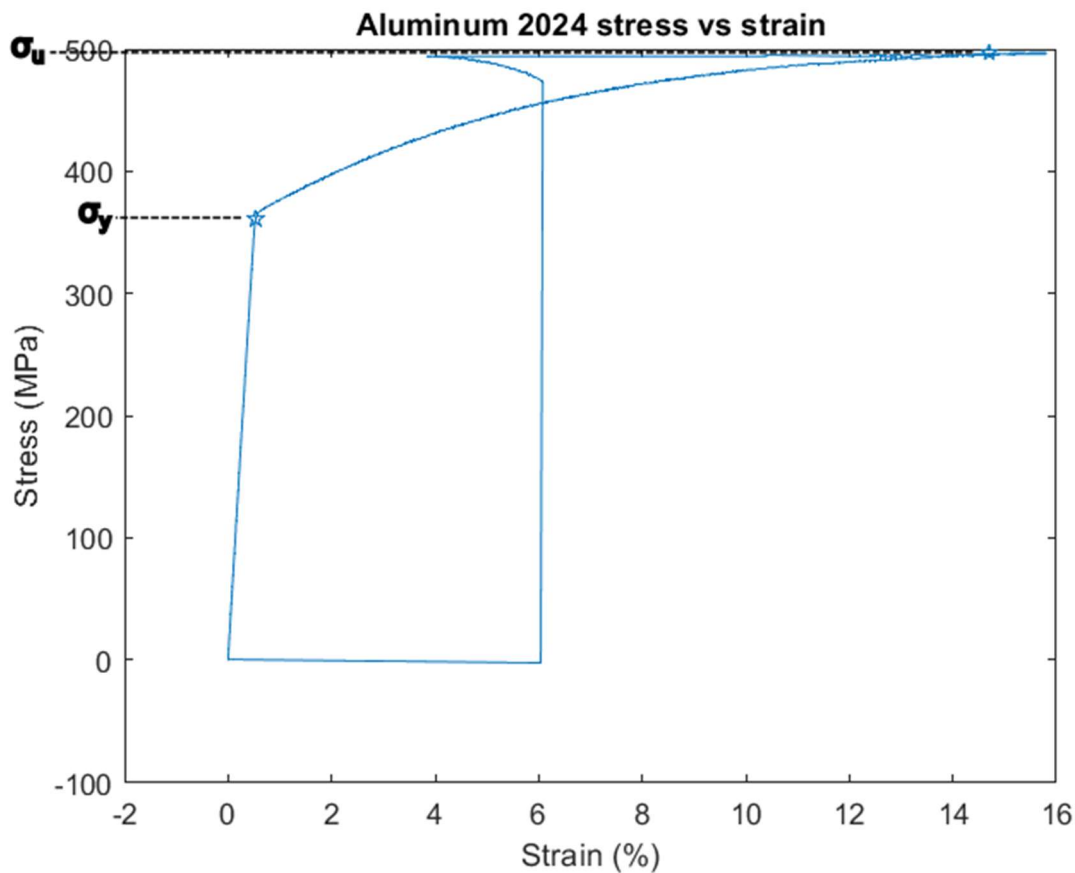


Figure 14: Stress-strain graph for aluminum 2024 with yield and ultimate stress

E (MPa)	693
σ_u (MPa)	497
σ_y (MPa)	361

Table 2: Al 2024 values from stress-strain curve

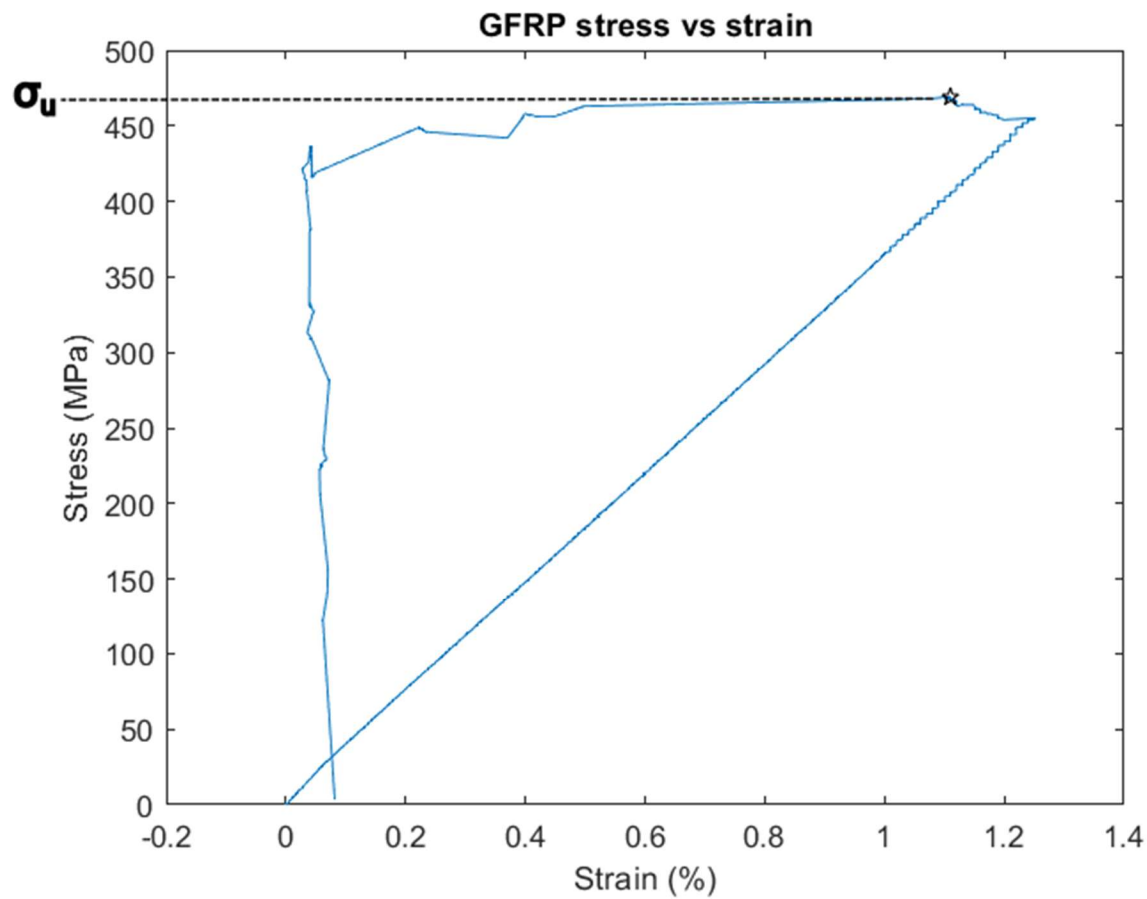


Figure 15: Stress-strain graph for GFRP

E (MPa)	373.9
σ_u (MPa)	469

Table 3: GFRP values from stress-strain curve

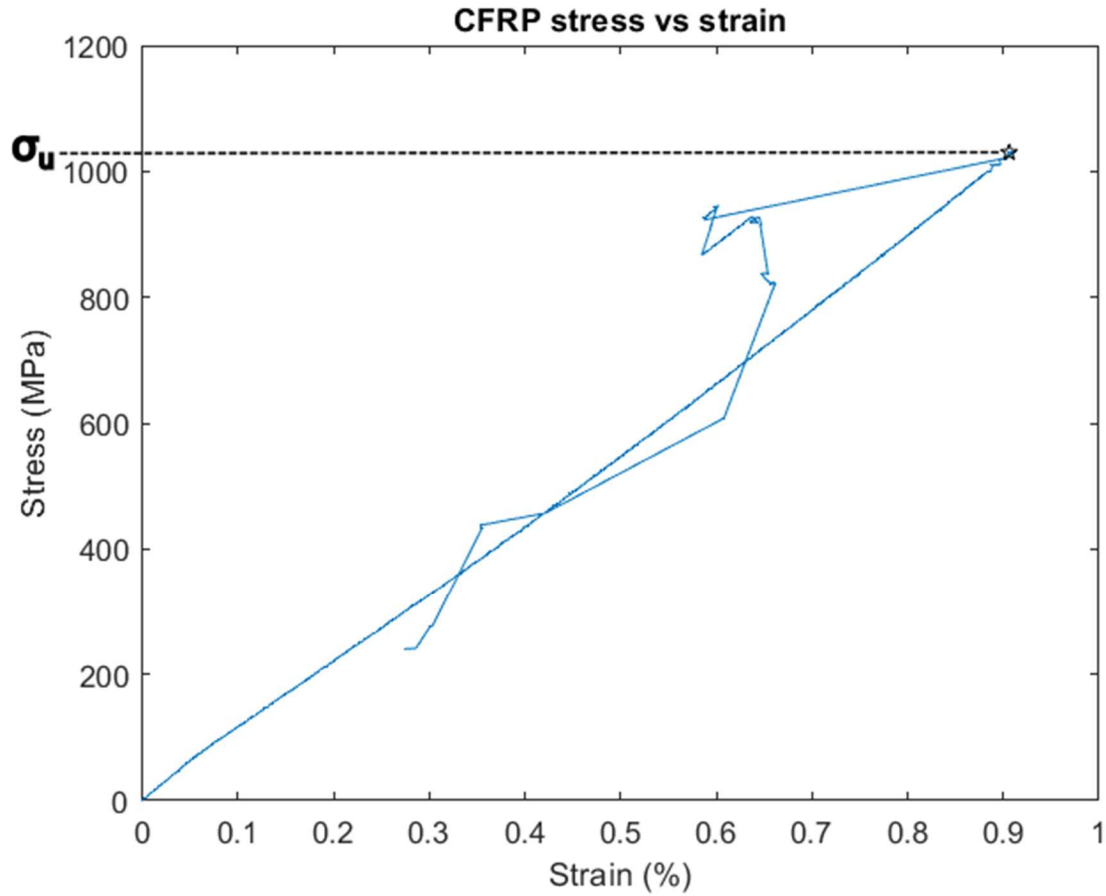


Figure 16: Stress-strain graph for CFRP

E (MPa)	1118.6
σ_u (MPa)	1030

Table 4: CFRP values from stress-strain curve

To check the validity of these values, they must be compared to the accepted values for the components found in Table 5.

Aluminum 2024	
E (MPa)	731
σ_u (MPa)	469
σ_y (MPa)	324
GFRP	
E (MPa)	276
σ_u (MPa)	N/A
CFRP	
E (MPa)	1378
σ_u (MPa)	99977

Table 5: Handbook values for the selected Al 2024, GFRP, CFRP

By using formula 2, the percent difference can be found for each value, found in Table 6.

$$p = \frac{\frac{ehandbook - xperiment}{handbook + experiment}}{2} * 100 \quad (2)$$

Aluminum 2024	
E (%)	1.326
σ_u (%)	1.449
σ_y (%)	2.7
GFRP	
E (%)	0.4067
σ_u (%)	N/A
CFRP	
E (%)	48.89
σ_u (%)	7.226

Table 5: Percent difference for the selected Al 2024, GFRP, CFRP vs handbook values

When comparing these materials, an important metric is the strength-to-weight comparison. With the testing materials, we specifically compared the ultimate tensile stress over the density. The higher the value, the more tensile stress the material can withstand compared to other materials at the same weight. In the experiment, Al 2024 had the lowest strength-to-weight ratio, while carbon fiber reinforced had the highest.

Aluminum 2024	
σ_u/ρ ((MPa*mm ³)/g)	178777
GFRP	
σ_u/ρ ((MPa*mm ³)/g)	250482
CFRP	
σ_u/ρ ((MPa*mm ³)/g)	717297

Table 6: Tested strength-to-weight values of Al 2024, GFRP, CFRP

For carbon fiber and other fiber-based composites, the Young's modulus can be calculated in an alternate way that it can be calculated, which is by looking at the compound under a microscope. In this, the sample image must be taken, then the fibers counted and measured for an average size. Then equation 3-5 can be used to find the modulus.

$$E = E_m * V_m + E_f * V_f \quad (3)$$

$$V_m = 1 - V_f \quad (4)$$

$$V_f = \frac{N\pi \text{ avg}^2}{LW} \quad (5)$$

Where:

E_m : Young's modulus for the matrix

V_m : Volume percent of the matrix

E_f : Young's modulus for the fiber

V_f : Volume percent of the fiber

N : Number of fibers

μ_{avg} : Average radius of fibers

L : Length of image

W : Width of image

With CFRP using figures 9-13, the values in table 7 were found, resulting in a Young's modulus of 126984 MPa, a 6% difference from the published data.

E_m (MPa)	3300
V_m (%)	0.4496
E_f (MPa)	228000
V_f (%)	0.5504
E (MPa)	126984
%dif E	5.9497

Table 7: Parameters along with their % difference

Conclusion

The results from the experiment show how tensile testing is a reliable method to obtain mechanical properties for certain materials while metallography is better for others, looking at our percent difference for E, we see values well within appropriate ranges for Aluminum-2024 and GFRP being around 1.3% and 0.4% this trend continuing for yield and ultimate stress, maxing out a 2.7% difference which was for the yield stress of the Aluminum-2024. However, CFRP shows a massive jump with almost a 50% error for the Young's Modulus and over 7% error for the ultimate stress when compared to published values. Recalculating the Young's Modulus for the CFRP sample using the rule of mixtures and metallography, we find a value of around 127000 MPa, which gives us a percent error of around 6%, a value much closer to the expected outcome. Some portion of the percent errors could be attributed to small disturbances or inaccuracies during the experiment, such as any inconsistencies with the samples. An easy solution would be to simply have more data for each type of sample. However, this explanation does not apply to the almost 50% error found on Young's Modulus for the CFRP, which entails either that the method used for calculating Young's Modulus for the other two samples does not work for CFRP, or that a critical error occurred during either testing or calculating. Considering that after recalculating Young's Modulus using a different method gave a value significantly closer to the expected, leads us to believe that CFRP's Young's Modulus cannot be calculated using the same method used on the Aluminum-2024 and GFRP samples.

Acknowledgements

This experiment was conducted as part of the AE-417 course at Embry-Riddle Aeronautical University. We would like to thank our lab instructor David Sypeck and teaching assistant Reshma Kubsad for providing the instructions and guidance for this experiment. Each team member contributed to the following sections of the lab report:

A. Bharadwaj: Introduction/Equipment

J. Nguyen Jr.: Abstract/Conclusion

R. Orr: Analysis

K. Satvaldy: Procedure/Results

Reference

Aluminum 2024-T4; 2024-T351. ASM material data sheet.

<https://asm.matweb.com/search/specificmaterial.asp?bassnum=ma2024t4>

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Hodgkinson, J. (2012). Testing the strength and stiffness of polymer matrix composites. In *Failure Mechanisms in Polymer Matrix Composites: Criteria, testing and industrial applications* (pp. 129–182). essay, Woodhead Publishing.

Holmes, Douglas. “Mechanics of Materials: Strain.” *Mechanics of Slender Structures RSS*, www.bu.edu/moss/mechanics-of-materials-strain.

Appendix

```
%{
Ryan Orr
AE417
Lab 3
%} clear clc

%% Aluminum
Al_data = readtable('Al_2024_metal_sample_tensile_test.csv');
Al_force = table2array(Al_data(:,1));

Al_position = table2array(Al_data(:,2));

Al_stress = table2array(Al_data(:,3));

Al_strain = table2array(Al_data(:,4));

E_Al_86 = Al_stress(86)/Al_strain(86);

E_Al_10 = Al_stress(10)/Al_strain(10);

E_Al_50 = Al_stress(50)/Al_strain(50);

E_Al_90 = Al_stress(90)/Al_strain(90);

E_Al_25 = Al_stress(25)/Al_strain(25); E_Al_average =
(E_Al_25+E_Al_90+E_Al_50+E_Al_10+E_Al_86)/5;

range_Al_E = E_Al_average-E_Al_average*.02;

%twopercent_strain = Al_strain(1:80)+.002;
%twopercent_stress = twopercent_strain*E_Al_average;
a=10; E_Al_test = 700; while
E_Al_test > range_Al_E

    E_Al_test = Al_stress(a)/Al_strain(a);

    Al_strain_yeild = Al_strain(a);
    Al_stress_yield = Al_stress(a);    a=a+1;
end

Al_stress_yield_location = find(Al_stress == Al_stress_yield);

Al_stress_yield_location = Al_stress_yield_location(1);
```

```

Al_stress_max = max(Al_stress); for b =
1:size(Al_stress)

    Al_stress_max_boolean = isequal(Al_stress(b),Al_stress_max); if
Al_stress_max_boolean == 1

        Al_stress_max_location = b;
Al_strain_maxStress_location = Al_strain(b);    return    end
end

%%%
figure("Name",'AL_stess_strain')
plot(Al_strain,Al_stress,'-p','MarkerIndices',[Al_stress_max_location,
Al_stress_yield_location]); hold on

%plot(twopercent_strain,twopercent_stress)
title('Aluminum 2024 stress vs strain'); xlabel('Strain
(%)'); ylabel('Stress (MPa)'); hold off

%%%
E_Al_average_accepted = 731; %MPa

Al_stress_yield_accepted = 324;

Al_stress_max_accepted = 469;

percdif_E_Al = (E_Al_average_accepted-E_Al_average)/
(E_Al_average_accepted+E_Al_average)/2 *100; % (%) percdif_stress_y_Al =
abs((Al_stress_yield_accepted-Al_stress_yield)/
(Al_stress_yield_accepted+Al_stress_yield)/2 *100); % (%) percdif_stress_u_Al =
abs((Al_stress_max_accepted-Al_stress_max)/
(Al_stress_max_accepted+Al_stress_max)/2 *100); % (%)

%%% d_Al =
8.58;

density_Al = 2.78/1000; %g/mm^3

Al_stress_max_density = Al_stress_max/density_Al; %Mpa*mm^3/g

%%% GFRP
GFRP_data = readtable("GFRP composite tensile_test.csv");
GFRP_force = table2array(GFRP_data(:,1));

```



```

GFRP_position = table2array(GFRP_data(:,2));
GFRP_stress = table2array(GFRP_data(:,3));
GFRP_strain = table2array(GFRP_data(:,4));

E_GFRP_10 = GFRP_stress(10)/GFRP_strain(10);
E_GFRP_100 = GFRP_stress(100)/GFRP_strain(100);
E_GFRP_200 = GFRP_stress(200)/GFRP_strain(200);
E_GFRP_300 = GFRP_stress(300)/GFRP_strain(300);
E_GFRP_500 = GFRP_stress(500)/GFRP_strain(500);
E_GFRP_800 = GFRP_stress(800)/GFRP_strain(800); E_GFRP_average =
(E_GFRP_200+E_GFRP_100+E_GFRP_300+E_GFRP_500+E_GFRP_800)/5;

GFRP_stress_max = max(GFRP_stress); for c =
1:size(GFRP_stress)

    GFRP_stress_max_boolean = isequal(GFRP_stress(c),GFRP_stress_max); if
GFRP_stress_max_boolean == 1

        GFRP_stress_max_location = c;

        GFRP_strain_maxStress_location = GFRP_strain(c);

    return end
end %%
figure('Name','GFRP_stess_strain')
plot(GFRP_strain,GFRP_stress,'p-','MarkerEdgeColor','black','MarkerIndices',
GFRP_stress_max_location) hold on title('GFRP stress vs strain'); xlabel('Strain (%)');
ylabel('Stress (MPa)'); hold off
%%
GFRP_stress_max_accepted = 276; %mpa
E_GFRP_average_accepted = 380;

percdif_E_GFRP = (E_GFRP_average_accepted-E_GFRP_average)/
(E_GFRP_average_accepted+E_GFRP_average)/2 *100; % (%) percdif_stress_u_GFRP =
abs((GFRP_stress_max_accepted-GFRP_stress_max)/
(GFRP_stress_max_accepted+GFRP_stress_max) *100); % (%)

%% d_GFRP = 9.53;
%mm r_GFRP =

```

```

d_GFRP/2; l_GFRP =
295; %mm m_GFRP =
39.4; %g

V_GFRP = pi*r_GFRP^2*l_GFRP;
density_GFRP = m_GFRP/V_GFRP; %g/mm^3

GFRP_stress_max_density = GFRP_stress_max/density_GFRP; %Mpa*mm^3/g

%% CFRP
CFRP_data = readtable("CFRP composite sample tensile test.csv");
CFRP_force = table2array(CFRP_data(:,1));

CFRP_position = table2array(CFRP_data(:,2));

CFRP_stress = table2array(CFRP_data(:,3));

CFRP_strain = table2array(CFRP_data(:,4));

E_CFRP_10 = CFRP_stress(10)/CFRP_strain(10);
E_CFRP_100 = CFRP_stress(100)/CFRP_strain(100);
E_CFRP_200 = CFRP_stress(200)/CFRP_strain(200);
E_CFRP_300 = CFRP_stress(300)/CFRP_strain(300);
E_CFRP_500 = CFRP_stress(500)/CFRP_strain(500);
E_CFRP_800 = CFRP_stress(800)/CFRP_strain(800); E_CFRP_average =
(E_CFRP_200+E_CFRP_100+E_CFRP_300+E_CFRP_500+E_CFRP_800)/5;

CFRP_stress_max = max(CFRP_stress); for d =
1:size(CFRP_stress)

    CFRP_stress_max_boolean = isequal(CFRP_stress(d),CFRP_stress_max);

    if CFRP_stress_max_boolean == 1

        CFRP_stress_max_location = d;
CFRP_strain_maxStress_location = CFRP_strain(d);    return    end
end %% figure("Name",'CFRP_stress_strain')

plot(CFRP_strain,CFRP_stress,'p-','MarkerEdgeColor','black','MarkerIndices',
CFRP_stress_max_location) hold on title('CFRP stress vs strain'); xlabel('Strain (%)');
ylabel('Stress (MPa)'); hold off

```

```

%%
CFRP_stress_max_accepted = 1378; %MPa

E_CFRP_average_accepted = 14.5*6.895*1000;%MPa

percdif_E_CFRP = (E_CFRP_average_accepted-E_CFRP_average)/
(E_CFRP_average_accepted+E_CFRP_average)/2 *100; % (%) percdif_stress_u_CFRP =
abs((CFRP_stress_max_accepted-CFRP_stress_max)/
(CFRP_stress_max_accepted+CFRP_stress_max)/2 *100); % (%)

%% d_CFRP = 9.384;
%mm r_CFRP =
d_CFRP/2; l_CFRP =
320; %mm m_CFRP =
31.78; %g

V_CFRP = pi*r_CFRP^2*l_CFRP; density_CFRP
= m_CFRP/V_CFRP; %g/mm^3

CFRP_stress_max_density = CFRP_stress_max/density_CFRP; %Mpa*mm^3/g

%%
E_m_CFRP = 3.3*1000; %MPa

E_f_CFRP = 228*1000; %MPa

V_f_CFRP_1 = (48*pi*((1.8+2+1.9+2.2+1.7+1.8)/12)^2)/(19.9*14);
V_f_CFRP_2 = (91*pi*((1.9+2+1.9+2.2+1.7+1.8)/12)^2)/(19.9*14);
V_f_CFRP_3 = (31*pi*((1.8+2+2.0+1.7+1.9+2)/12)^2)/(19.9*14);
V_f_CFRP_4 = (24*pi*((1.8+2+2.0+1.7+1.9+1.8)/12)^2)/(19.9*14);
V_f_CFRP_5 = (65*pi*((1.8+2+2.9+1.7+1.9+2)/12)^2)/(19.9*14);
V_f_CFRP_av = (V_f_CFRP_1+V_f_CFRP_2+V_f_CFRP_3+V_f_CFRP_4+V_f_CFRP_5)/5;
V_m_CFRP = 1-V_f_CFRP_av; E_CRFP_calc =
E_m_CFRP*V_m_CFRP+E_f_CFRP*V_f_CFRP_av;

percdif_E_CFRP_calc = abs(E_CFRP_average_accepted-E_CRFP_calc)/
(E_CFRP_average_accepted+E_CRFP_calc)/2 *100; % (%)

```